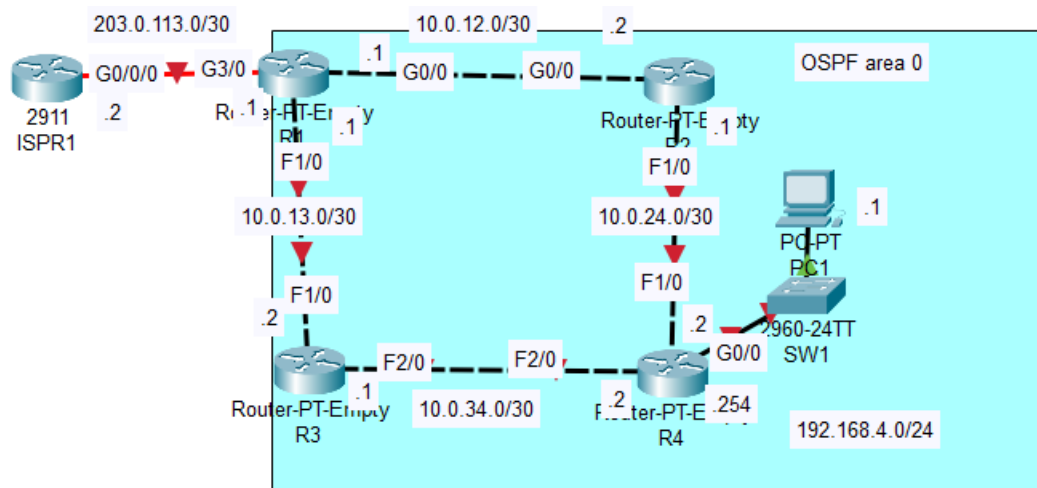




Topology

1. Configure the appropriate hostnames and IP addresses on each device. Enable router interfaces. (You don't have to configure ISPR1)

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#int g0/0
R1(config-if)#ip address 10.0.12.1 255.255.255.252
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

R1(config-if)#int f1/0
R1(config-if)#ip address 10.0.13.1 255.255.255.252
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

R1(config-if)#int g3/0
R1(config-if)#ip address 203.0.113.1 255.255.255.252
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet3/0, changed state to up
```

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#int g0/0
R2(config-if)#ip address 10.0.12.2 255.255.255.252
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R2(config-if)#int f1/0
R2(config-if)#ip address 10.0.24.1 255.255.255.252
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R3
R3(config)#int f1/0
R3(config-if)#ip address 10.0.13.2 255.255.255.252
R3(config-if)#no shutdown

R3(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

R3(config-if)#int f2/0
R3(config-if)#ip address 10.0.34.1 255.255.255.252
R3(config-if)#no shutdown

R3(config-if)#
%LINK-5-CHANGED: Interface FastEthernet2/0, changed state to up
```

```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R4
R4(config)#ip address 10.0.34.2 255.255.255.252
^
% Invalid input detected at '^' marker.

R4(config)#int f2/0
R4(config-if)#ip address 10.0.34.2 255.255.255.252
R4(config-if)#int f1/0
R4(config-if)#ip address 10.0.24.2 255.255.255.252
R4(config-if)#no shutdown

R4(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

R4(config-if)#int f2/0
R4(config-if)#no shutdown

R4(config-if)#
%LINK-5-CHANGED: Interface FastEthernet2/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up

R4(config-if)#int g0/0
R4(config-if)#192.168.4.2 255.255.255.0
^
% Invalid input detected at '^' marker.

R4(config-if)#ip address 192.168.4.2 255.255.255.0
R4(config-if)#no shutdown

R4(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

```

2. Configure a loopback interface on each router (1.1.1.1/32 for R1, 2.2.2.2/32 for R2, etc.)

```

R1(config)#int 10

R1(config-if)#
%LINK-5-CHANGED: Interface Loopback0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

R1(config-if)#ip address 1.1.1.1 255.255.255.255
R2(config)#int 10

R2(config-if)#
%LINK-5-CHANGED: Interface Loopback0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

R2(config-if)#ip address 2.2.2.2 255.255.255.255

```

```

R3(config)#int 10

R3(config-if)#
%LINK-5-CHANGED: Interface Loopback0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

R3(config-if)#ip address 3.3.3.3 255.255.255.255
R4(config)#int 10

R4(config-if)#
%LINK-5-CHANGED: Interface Loopback0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

R4(config-if)#ip address 4.4.4.4 255.255.255.255

```

3. Enable OSPF directly on each interface of the routers. Configure passive interfaces as appropriate.

```

R1(config)#router ospf 1
R1(config-router)#network 10.0.12.0 0.0.0.3 area 0
R1(config-router)#network 10.0.13.0 0.0.0.3 area 0
R1(config-router)#network 1.1.1.1 0.0.0.0 area 0

R1(config-router)#passive-interface 10

R2(config)#router ospf 2
R2(config-router)#network 10.0.0.0 0.0.255.255 area 0
R2(config-router)#
23:55:44: %OSPF-5-ADJCHG: Process 2, Nbr 1.1.1.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
n
23:55:45: %OSPF-5-ADJCHG: Process 2, Nbr 4.4.4.4 on FastEthernet1/0 from LOADING to FULL, Loading Done
etwork 2.2.2.2 0.0.0.0 area 0
R2(config-router)#passive-interface 10

R3(config)#router ospf 3
R3(config-router)#network 10.0.13.2 0.0.0.0 area 0
R3(config-router)#network 10.0.34.1 0.0.0.0
23:49:35: %OSPF-5-ADJCHG: Process 3, Nbr 1.1.1.1 on FastEthernet1/0 from LOADING to FULL, Loading Done
area 0
R3(config-router)#
23:51:02: %OSPF-5-ADJCHG: Process 3, Nbr 4.4.4.4 on FastEthernet2/0 from LOADING to FULL, Loading Done

R3(config-router)#network 3.3.3.3 0.0.0.0 area 0
R3(config-router)#passive-interface 10
R4(config)#router ospf 4
R4(config-router)#network 0.0.0.0 255.255.255.255 area 0
R4(config-router)#
23:51:02: %OSPF-5-ADJCHG: Process 4, Nbr 3.3.3.3 on FastEthernet2/0 from LOADING to FULL, Loading Done

R4(config-router)#passive-interface g0/0
R4(config-router)#passive-interface 10
Configure ospf on all interfaces

```

4. Configure R1 as an ASBR that advertises a default route in to the OSPF domain.

```
R1>en
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#default-information originate
R1(config-router)#exit
R1(config)#ip route 0.0.0.0 0.0.0.0 203.0.113.2
```